CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology

LAB Exercise II –

Research Gap Identification.

Date of assignment: 23-09-2026
Date of submission: 03-10-2026

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Domain :

Marine Environmental Monitoring & Coastal Ecosystem-Preservation

Lab 2

Lab Exercise II:
Research Gap Identification.

1. Literature Summary and Findings

The five foundational papers reviewed in Lab 1 establish the operational boundaries and core physical limitations of marine plastic debris monitoring across laboratory, numerical, and satellite platforms. Guffogg et al. (2021) demonstrated via in-situ hyperspectral radiometry that only two polymer absorption features (1215 nm and 1732 nm) remain observable against natural quartz and carbonate sand substrates, while highlighting that fractional surface coverage does not scale linearly with diagnostic absorption depth. Wichmann et al. (2019) and van Sebille et al. (2020) demonstrated that ocean circulation advects a substantial proportion of microplastics beneath the surface layer and into high-latitude waters, revealing that satellites observe only the visible surface fraction of the total debris budget. Kikaki et al. (2022) and Biermann et al. (2020) demonstrated the operational utility of Sentinel-2 MSI data, presenting the benchmark MARIDA dataset and the Floating Debris Index (FDI) respectively. Overall, the literature confirms that while optical marine litter monitoring has advanced, three major bottlenecks persist: high spectral confusion between plastics and natural flotsam (driftwood/vegetation), optical signal degradation in turbid coastal waters, and the complete absence of transport-model coupling to account for subsurface drift.

2. Five Issues Identified in the Existing Literature

Issue 1: High false-positive classification between weathered polymers and natural organic flotsam

What is missing / could be improved: Both Guffogg et al. and the MARIDA dataset (Kikaki et al.) reveal that weathered plastics, decomposing Sargassum, and driftwood exhibit nearly identical baseline red-edge slopes in 10-meter and 20-meter multispectral data, causing high false-alarm rates in standard pixel classifiers.

New questions that arise: Can high-order spatial texture co-occurrence metrics (GLCM contrast and homogeneity) across shortwave infrared (SWIR) bands reliably separate rigid, synthetic polymer geometries from cellular organic flotsam?

How my research could contribute: I can construct a feature-engineering pipeline combining spectral indices with texture features across Sentinel-2 bands to improve discrimination between debris and natural flotsam using the MARIDA benchmark.

Future direction suggested by the authors: Kikaki et al. explicitly recommend incorporating multi-scale spatial texture descriptors and directional spatial filters to separate anthropogenic polymers from natural flotsam.

Issue 2: Non-linear optical masking and lack of sub-pixel plastic abundance quantification

What is missing / could be improved: Guffogg et al. identified that fractional plastic surface coverage fails to maintain a linear relationship with absorption band depth when mixed with wet substrate, which leaves current methods unable to quantify plastic mass or density per pixel.

New questions that arise: Can a non-linear spectral unmixing model or polynomial calibration function estimate fractional plastic coverage below 10% within a mixed coastal pixel?

How my research could contribute: I can design an empirical unmixing workflow that isolates polymer endmember contributions from background sand and foam radiance.

Future direction suggested by the authors: Guffogg et al. recommend conducting multi-concentration field and flume experiments to model composite reflectance changes across varying substrate moisture levels.

Issue 3: Complete decoupling between optical surface satellite detections and hydrodynamic transport physics

What is missing / could be improved: Wichmann et al. and van Sebille et al. confirm that ocean currents and vertical mixing move massive amounts of debris beneath the surface layer, yet satellite remote-sensing studies operate strictly on instantaneous surface snapshots without physical vector validation.

New questions that arise: Can hydrodynamic Lagrangian velocity fields be statistically coupled with satellite surface detections to estimate potential subsurface accumulation corridors?

How my research could contribute: My research can correlate spatial surface detections from satellite imagery with Copernicus hydrodynamic current velocity grids to map coastal convergence and trajectory trends.

Future direction suggested by the authors: van Sebille et al. emphasize the need to bridge the gap between physical ocean transport simulations and remote sensing observations.

Issue 4: Lack of cross-regional domain generalization in machine-learning debris models

What is missing / could be improved: Existing benchmarks such as MARIDA and Biermann et al.'s models are trained on narrow geographic regions, causing classification performance to collapse when applied to uncalibrated coastal sites with different optical water types.

New questions that arise: How significantly does classification accuracy degrade when transferring a model from clear Mediterranean waters to sediment-rich tropical coastal zones?

How my research could contribute: I can evaluate pre-trained Sentinel-2 classification pipelines across geographically disjoint coastal zones to quantify the exact domain-shift error.

Future direction suggested by the authors: Kikaki et al. highlight the necessity of incorporating geographically diverse, verified open-source scenes into future benchmark repositories.

Issue 5: Severe spectral masking and failure of optical indices in turbid coastal and estuarine discharge zones

What is missing / could be improved: Biermann et al. documented that the Floating Debris Index (FDI) depends on clear ocean water baselines and degrades severely in turbid river mouths, where high concentrations of Suspended Particulate Matter (SPM) elevate Red and NIR backscatter and obscure polymer absorption anomalies.

New questions that arise: Can an adaptive baseline correction be formulated to decouple sediment backscatter from floating polymer signatures in high-turbidity coastal boundaries?

How my research could contribute: I can evaluate spectral index performance across variable turbidity conditions to establish a sediment-compensated detection baseline for river discharge plumes.

Future direction suggested by the authors: Biermann et al. explicitly identify adapting optical debris detection pipelines to sediment-heavy, complex coastal and inland waterways as an essential open challenge.